



TURNING WASTE INTO WEALTH: Karnataka's ₹20,000–Crore Circular Economy Opportunity

Karnataka stands at a critical crossroads where rapid urbanisation, industrial growth, and rising consumption are generating unprecedented volumes of waste.

What is often viewed as an environmental burden, however, holds the potential to become one of the state's strongest economic drivers. If fully harnessed, Karnataka's waste economy could unlock a circular economy worth 15,000–20,000 crore annually while generating 1–2 lakh jobs and reduce emissions by up to 25%.

Urban Karnataka generates nearly 13,000 tonnes of municipal solid waste (MSW) every day, with Bengaluru alone contributing about 6,000 tonnes. Around 50–60% is organic, 15–25% recyclable dry waste, and 20–30% inert material, including construction debris and silt.

Efficient waste management can produce compost, bio-fertilizers, compressed biogas, refuse-derived fuel (RDF), and recoverable plastics, metals, and paper.

Plastic waste presents another major opportunity. Karnataka generates nearly 3.6 lakh tonnes annually. Properly processed, plastic waste can be converted into recycled granules, road construction material, and RDF for cement plants.

Karnataka also generates around 2.9 lakh tonnes of e-waste annually, valued at nearly ₹2,886 crore. Yet only 10–15% is formally recovered. E-waste contains valuable metals such as gold, silver, copper, and rare earth elements critical for modern

technologies. Developing a robust urban mining ecosystem could unlock over ₹2,500 crore annually while reducing dependence on imported raw materials.

Industrial and hazardous waste adds another dimension. Karnataka produces nearly 9.17 lakh tonnes annually, most of which is managed scientifically. Through co-processing and recovery of chemicals and solvents, this segment can generate ₹1,000–2,000 crore annually while lowering environmental risks.

Biomedical waste, though smaller in volume at about 74 tonnes per day, supports a growing compliance-driven service economy worth an estimated ₹100–300 crore annually, alongside opportunities for energy recovery.

Construction and demolition (C&D) waste, accounting for 15–25% of urban waste, is another underutilised resource. Processed properly, it can produce M-sand, aggregates, bricks, and paver blocks, reducing dependence on natural sand mining. The market potential for recycled construction material is estimated at ₹2,000–4,000 crore annually.

Organic and agricultural waste, which forms the bulk of municipal waste, offers immense scope for circular solutions. Through composting, vermicomposting, and bioenergy generation, this segment can generate ₹1,500–3,000 crore annually while improving soil health and reducing dependence on chemical fertilizers.

Sewage and wastewater are perhaps the most overlooked



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resources. Treated wastewater can be reused in industry, construction, and landscaping, while sewage sludge can be converted into biogas and electricity.

The way forward requires a shift from waste management to resource management. Strengthening segregation at source, expanding recycling infrastructure, creating markets for secondary materials, and integrating the informal sector are critical. Policy support, technological innovation, and private sector participation will be equally important.

With its strong industrial base, active startup ecosystem, and progressive urban governance, Karnataka is well positioned to transform its waste challenge into a model circular economy that addresses environmental concerns while driving economic growth. The message is clear: waste is not a liability, but a multi-sector resource waiting to be unlocked.